

I. Fluid Ontology

II. Fiedler-Markov QBE

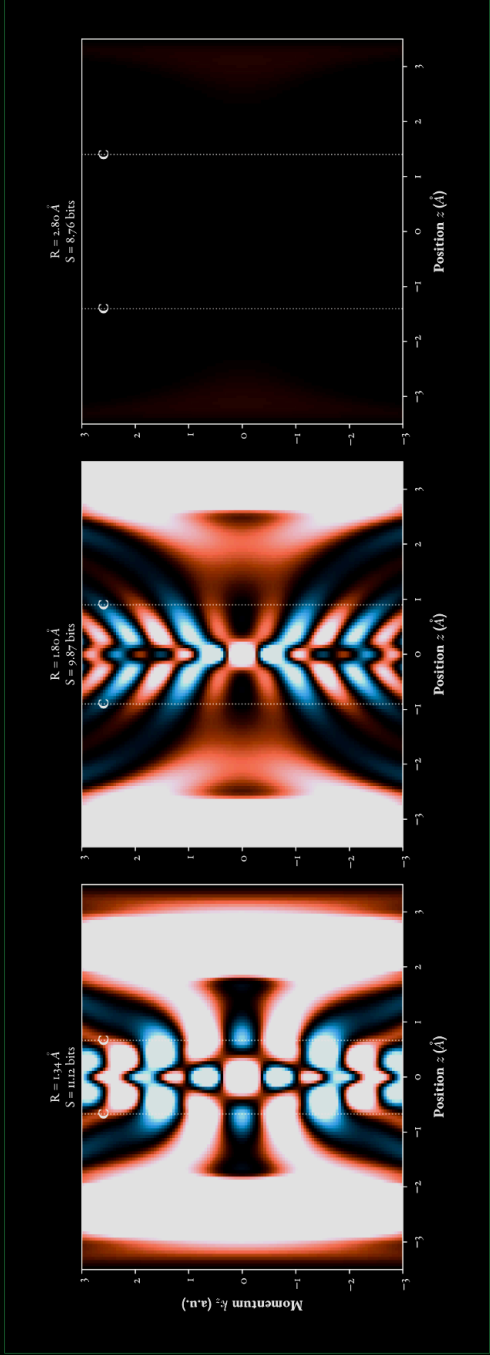
III. Conical Intersections

IV. Active Space Replacement

MATHEMATICAL ENGINE // COLLISION OPERATOR

Quantum Boltzmann Kinetic Engine

Bridging kinetic theory and matrix mechanics via trace-preserving matrix propagation.



Autonomous BGK Relaxation

Density matrices rotate toward a Fermi-Dirac equilibrium via a discrete Bhatnagar-Gross-Krook loop.

Fiedler Friction

Collision frequency is dynamically derived from the algebraic connectivity of the HOMO-LUMO graph.

Thermodynamic Valve

Fractional mixing channels open strictly at topological singularities.

CORE PARADIGM SHIFT

Moat: Recovering exact Hartree-Fock stationarity at the zero-temperature limit.